Conjugate Heat Transfer Setup

1. Domain Architecture and Multi-Physics Coupling

A three-dimensional finite element formulation is established to solve coupled turbulent fluid dynamics and conjugate heat transfer in a liquid-cooled capacitor assembly. The computational domain $\Omega = \Omega_s \cup \Omega_f$ is partitioned into internal solid conducting sub-domains (Ω_s) and the surrounding coolant fluid domain (Ω_f).

Taking advantage of geometric and thermofluidic symmetry, a half-symmetry domain is modeled across a longitudinal plane Γ_{sym} cutting through the core and flow channel. This reduction halves the computational degrees of freedom (DOF) while permitting dense near-wall boundary layer resolution required to capture steep viscous and thermal gradients.

2. Governing Field Equations Solved by Finite Elements

The formulation couples the incompressible Reynolds-Averaged Navier–Stokes (RANS) equations closed with the Realizable k–ε turbulence model to the convective-diffusive thermal energy equations across solid and liquid domains.

2.1 Turbulent Fluid Flow in Ω_f (RANS with Realizable k–ε Turbulence)

The coolant velocity and pressure fields are governed by mass and momentum conservation:

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\rho_f(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[(\mu_f + \mu_T)\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right)\right] + \rho_f \mathbf{g} \tag{2}$$

where $\mathbf{u}$ is the fluid velocity vector, p is static pressure, ρ_f is fluid density, μ_f is dynamic molecular viscosity, $\mathbf{g}$ is gravitational acceleration, and μ_T is turbulent eddy viscosity.

Turbulence closure is provided by the **Realizable k–ε model**, which enforces mathematical realizability constraints on normal Reynolds stresses ($\overline{u_i^2} \geq 0$) and yields superior performance for channel boundary layers and separating flows:

$$\rho_f(\mathbf{u} \cdot \nabla)k = \nabla \cdot \left[\left(\mu_f + \frac{\mu_T}{\sigma_k}\right)\nabla k\right] + P_k - \rho_f \varepsilon \tag{3}$$

$$\rho_f(\mathbf{u} \cdot \nabla)\varepsilon = \nabla \cdot \left[\left(\mu_f + \frac{\mu_T}{\sigma_\varepsilon}\right)\nabla \varepsilon\right] + \rho_f C_1 S \varepsilon - \rho_f C_2 \frac{\varepsilon^2}{k + \sqrt{\nu_f \varepsilon}} \tag{4}$$

where k is turbulent kinetic energy, ε is turbulent dissipation rate, and $\nu_f = \mu_f/\rho_f$ is kinematic viscosity. The turbulent kinetic energy production rate is $P_k = \mu_T S^2$, where $S = \sqrt{2\mathbf{S} : \mathbf{S}}$ is the modulus of the mean strain rate tensor $\mathbf{S} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$. The eddy viscosity μ_T is defined with a dynamic, flow-dependent coefficient C_μ:

$$\mu_T = \rho_f C_\mu \frac{k^2}{\varepsilon}, \quad C_\mu = \frac{1}{A_0 + A_s \frac{kU^*}{\varepsilon}}, \quad U^* \equiv \sqrt{\mathbf{S} : \mathbf{S} + \mathbf{\Omega} : \mathbf{\Omega}} \tag{5}$$

where $\mathbf{\Omega}$ is the mean rotation rate tensor. The empirical model closure parameters are:

$$C_1 = \max\left(0.43, \frac{\eta}{\eta + 5}\right), \quad \eta = S\frac{k}{\varepsilon}, \quad C_2 = 1.9, \quad \sigma_k = 1.0, \quad \sigma_\varepsilon = 1.2, \quad A_0 = 4.04 \tag{6}$$

2.2 Conjugate Heat Transfer Formulation (Energy Conservation)

The thermal energy equation governs conduction across solid sub-domains and coupled advection-diffusion within the fluid:

1. **Solid Domains (Ω_s):** Heat conduction with internal volumetric heat dissipation:

$$\rho_s C_{p,s} \frac{\partial T}{\partial t} = \nabla \cdot (\mathbf{k}_s \cdot \nabla T) + \dot{q}_v \tag{7}$$

   where ρ_s and $C_{p,s}$ are density and specific heat capacity of the respective solid components, and $\dot{q}_v$ is the volumetric heat generation rate. For the wound capacitor core, an orthotropic thermal conductivity tensor is prescribed:

$$\mathbf{k}_{\text{core}} = \text{diag}(k_\parallel, k_\perp, k_\perp) \tag{8}$$

   capturing high in-plane thermal conduction along continuous metallization layers ($k_\parallel$) versus cross-layer contact resistance through stacked polymer films ($k_\perp$).

2. **Coolant Fluid Domain (Ω_f):** Convective-diffusive heat transport:

$$\rho_f C_{p,f}\left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T\right) = \nabla \cdot ((k_f + k_T)\nabla T) \tag{9}$$

   where k_f is the fluid molecular thermal conductivity, and k_T is the turbulent thermal conductivity evaluated via the Reynolds analogy:

$$k_T = \frac{C_{p,f}\mu_T}{\text{Pr}_T} \tag{10}$$

   with Pr_T denoting the turbulent Prandtl number.

3. Boundary and Interface Conditions

The boundary conditions governing the coupled hydrodynamic and thermal domains are specified as follows:

- **Fluid Inlet (Γ_{in}):** Prescribed mass flow rate condition $\int_{\Gamma_{\text{in}}} \rho_f(\mathbf{u}\cdot\mathbf{n})\,dA = \dot{m}_{\text{in}}$, along with a Dirichlet inflow temperature $T\big|_{\Gamma_{\text{in}}} = T_{\text{in}}$, turbulent intensity I_T, and turbulent length scale L_T.
- **Fluid Outlet (Γ_{out}):** Static gauge pressure outlet $p\big|_{\Gamma_{\text{out}}} = 0$, $\left[(\mu_f + \mu_T)\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right)\right]\cdot\mathbf{n} = \mathbf{0}$, with convective thermal outflow $-\mathbf{n}\cdot(k_f\nabla T) = 0$.
- **Solid-Fluid Interface (Γ_{int}):** Strict Dirichlet–Neumann continuity across all wetting boundaries:

$$T_s = T_f, \quad -\mathbf{n}\cdot(\mathbf{k}_s\cdot\nabla T_s) = -\mathbf{n}\cdot((k_f + k_T)\nabla T_f) \tag{11}$$

- **Channel Wall Boundaries (Γ_{wall}):** No-slip velocity conditions ($\mathbf{u} = \mathbf{0}$) enforced using scalable wall functions.
- **Symmetry Plane (Γ_{sym}):** Zero normal velocity, zero tangential traction, and adiabatic thermal isolation:

$$\mathbf{u}\cdot\mathbf{n} = 0, \quad \nabla k\cdot\mathbf{n} = 0, \quad \nabla\varepsilon\cdot\mathbf{n} = 0, \quad -\mathbf{n}\cdot\mathbf{q} = 0 \tag{12}$$

- **Core Heat Source (Ω_{core}):** Volumetric heat dissipation representing internal dielectric and Joule losses: $\int_{\Omega_{\text{core}}} \dot{q}_v\,dV = P_{\text{core}}$.
- **External Housing Boundaries (Γ_{ext}):** Mixed Robin condition accounting for ambient convection and radiation:

$$-\mathbf{n}\cdot\mathbf{q} = h_{\text{ext}}(T - T_{\text{ext}}) + \varepsilon_{\text{rad}}\sigma_{\text{SB}}(T^4 - T_{\text{amb}}^4) \tag{13}$$

4. Weak Formulation and Numerical Stabilization

In the finite element discretization, the governing PDEs are multiplied by test functions $(\mathbf{v}, q, \psi)$ and integrated over element volumes:

$$\int_{\Omega_f} \rho_f(\mathbf{u}\cdot\nabla\mathbf{u})\cdot\mathbf{v}\,d\Omega - \int_{\Omega_f} p(\nabla\cdot\mathbf{v})\,d\Omega + \int_{\Omega_f} (\mu_f + \mu_T)\nabla\mathbf{u} : \nabla\mathbf{v}\,d\Omega = \int_{\Omega_f} \rho_f\mathbf{g}\cdot\mathbf{v}\,d\Omega \tag{14}$$

$$\int_{\Omega} \rho C_p(\mathbf{u}\cdot\nabla T)\psi\,d\Omega + \int_{\Omega} (\mathbf{k}\cdot\nabla T)\cdot\nabla\psi\,d\Omega = \int_{\Omega_{\text{core}}} \dot{q}_v\psi\,d\Omega - \int_{\Gamma_{\text{ext}}} q_{\text{ext}}\psi\,d\Gamma \tag{15}$$

4.1 Numerical Stabilization: SUPG and PSPG Schemes

To suppress non-physical oscillations in advection-dominated high-Péclet regimes and permit equal-order linear (P_1–P_1) velocity-pressure interpolation without violating the Ladyzhenskaya–Babuška–Brezzi (LBB) condition, weighted residual stabilization terms are added:

$$\sum_e \int_{\Omega_e} \left[\tau_{\text{SUPG}}(\mathbf{u}\cdot\nabla\mathbf{v}) + \tau_{\text{PSPG}}\frac{\nabla q}{\rho_f}\right]\cdot\mathcal{R}_{\text{mom}}\,d\Omega_e + \sum_e \int_{\Omega_e} \tau_{\text{th}}(\mathbf{u}\cdot\nabla\psi)\mathcal{R}_{\text{energy}}\,d\Omega_e \tag{16}$$

where $\mathcal{R}_{\text{mom}}$ and $\mathcal{R}_{\text{energy}}$ represent momentum and energy residuals, and τ denotes element relaxation parameters.

4.2 Operator Coercivity and Diffusion Singularity

The second-order elliptic diffusion operator in the fluid thermal equation generates the diffusive element stiffness matrix $K_{ij}^e = \int_{\Omega_e} k_f\nabla\phi_i\cdot\nabla\phi_j\,d\Omega$. If the fluid thermal conductivity is zero ($k_f = 0$), K_{ij}^e vanishes identically. The energy equation loses its elliptic nature and degenerates into an un-dissipated first-order hyperbolic equation. In addition, the interface flux condition forces an artificial adiabatic boundary onto the solid while the fluid cannot conduct heat across the boundary, rendering the global Jacobian singular and causing solver divergence. Enforcing $k_f > 0$ restores bilinear form coercivity and enables robust quadratic convergence.

5. Solver Architecture and Solution Strategy

The multi-physics system is solved using a **Segregated Solver Algorithm**:

- **Segregated Group 1 (Hydrodynamics):** Solves the non-linear velocity, pressure, k, and ε fields using GMRES with multigrid preconditioning.
- **Segregated Group 2 (Energy Conservation):** Solves the conjugate temperature field T across solids and fluid using the direct PARDISO solver, ensuring absolute stability under anisotropic core conduction.
- **Study Sequences:** Damped Newton iterations for steady-state operating equilibrium, and implicit Backward Differentiation Formula (BDF) time-stepping for transient thermal history tracking.